

 poolside/**Laguna-S-2.1-INT4** 

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⋮

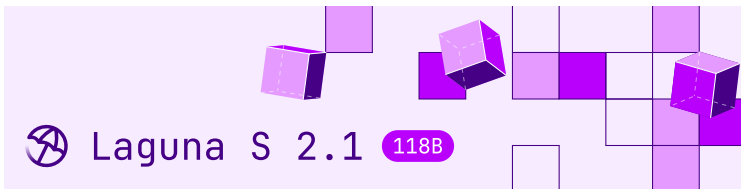
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 **Model card**

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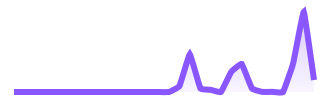
Laguna S 2.1-INT4

Laguna S 2.1-INT4 is a 117.6B total parameter Mixture-of-Experts model with 8.5B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 36 out of 48 layers for fast inference and low KV cache requirements.

Highlights

- Mixed SWA and global attention layout:**
 Laguna S 2.1 uses softplus gating with per-layer rotary scales, enabling mixed SWA (Sliding

Downloads last month
21,139



 **Safetensors** 

Model size 121B params

Tensor type F32 · BF16 · I32

 Chat template

 Files info

 **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.



Ask for provider support

 **Model tree for poolside/Laguna-S...**

Base model poolside/Laguna-S-2.1

 Quantized (56) [this model](#)

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Laguna S 2.1  Collection

Window Attention) and global attention layers in a 3:1 ratio (across 48 total layers)

- **KV cache in FP8:** KV cache quantized to FP8, reducing memory per token
- **Native reasoning support:** Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
- **Local-ready:** At 117.6B total parameters and 8.5B activated, the INT4 weights are roughly 72 GB. Available on [Ollama](#) and [llama.cpp](#) (BF16 and Q4_K_M only)
- **OpenMDW-1.1 license:** Use and modify the model and associated materials freely for commercial and non-commercial purposes ([learn more about OpenMDW](#))

Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 117.6B total with 8.5B activated per token
- Optimizer: Muon
- Layers: 48 layers (12 layers with global attention, 36 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert
- Sliding Window: 512 tokens
- Modality: text-to-text
- Context window: 262,144 tokens

- Reasoning support: interleaved thinking with preserved thinking

Benchmark results



Model	Size	Terminal-Bench 2.1	SWE-bench Multilingual	SWE Bench Pro (Public Dataset)
Laguna S 2.1	118B-A8B	70.2%	78.5%	59.4%

Model	Size	Terminal-Bench 2.1	SWE-bench Multilingual	SWE Bench Pro (Public Dataset)
Tencent Hy3	295B-A21B	71.7%	75.8%	57.9%
Inkling	975B-A41B	63.8%	-	54.3%
Nemotron 3 Ultra	550B-A55B	56.4%	67.7%	-
DeepSeek-V4-Pro Max	1.6T-A49B	64.0%*	76.2%	55.4%
Kimi K3	2800B-A50B	88.3%	-	-
Qwen 3.7 Max	-	74.5%*	78.3%	60.6%
Muse Spark 1.1	-	80%	-	61.5%
Claude Fable 5	-	88%	-	80.3%

Benchmarks as of 21 July 2026. Laguna S 2.1 in **bold**; a dash (-) marks a benchmark a model was not evaluated on. Scores marked * are as reported by third parties: Terminal-Bench 2.1 and DeepSWE via Artificial Analysis, SWE Atlas via Scale AI's official leaderboard, and Toolathlon Verified via its official leaderboard. Full evaluation trajectories: trajectories.poolside.ai.

Context length

This checkpoint ships configured for a 262,144-token (256K) context window. This is the configuration we recommend for best output quality.

The weights are native 1M checkpoints: training included a long-context extension stage up to 1,048,576 tokens, and quantization was calibrated at the 1M configuration. If you need more than 256K of context, restore the 1M configuration by editing `config.json`:

```
"rope_parameters": {  
  "full_attention": {  
    "factor": 128.0,  
    "attention_factor": 1.4852030263919618  
  }  
},  
"max_position_embeddings": 1048576
```

You may experience quality degradation with the 1M configuration. If you use it, we recommend sampling with `temperature <= 0.7` and `top_p <= 0.95`.

Recommended sampling

For the best balance of quality and reliability we recommend sampling with `temperature 0.7` and `top_p 0.95`.

Local deployment

Laguna S 2.1-INT4 is supported in vLLM and Transformers. Use Laguna S 2.1 with Ollama (with

MLX support) or Llama.cpp (BF16 and Q4_K_M only) for the best results on your local machine.

vLLM

“Serving requires vLLM 0.25.0 or later. See the main [Laguna S 2.1 model card](#) for the full recipe.”

The full vLLM recipe is on the main [Laguna S 2.1 model card](#) and on the [vLLM recipes page](#).

Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-S-2.1-INT4` substituted for the model ID. No extra flags required.

INT4 support landed in vLLM with [vllm-project/vllm#47154](#). This checkpoint mixes INT4 and group-quantized INT8 experts, which earlier builds rejected in the Marlin WNA16 MoE path. Use a vLLM build that includes this fix.

Optional: speculative decoding with DFlash. Pair with the quantization-matched draft model [poolside/Laguna-S-2.1-DFlash-INT4](#) by adding `--speculative-config '{\"model\": \"poolside/Laguna-S-2.1-DFlash-INT4\", \"num_speculative_tokens\": 15, \"method\": \"dflash\"}'` to the [serve command](#).

SGLang

The full SGLang recipe is on the main [Laguna S 2.1 model card](#) and on the [SGLang cookbook](#). SGLang reads the checkpoint `quantization_config`, so you can use the same launch command after replacing the model ID

Transformers

The full Transformers recipe is on the main [Laguna S 2.1 model card](#). Substitute poolside/Laguna-S-2.1-INT4 for the model ID; quantization is detected automatically from `quantization_config`.

llama.cpp

Serve the GGUF builds with Poolside's llama.cpp fork, branch [laguna](#). See the main [Laguna S 2.1 model card](#) for the build recipe. llama.cpp serves the BF16, Q8_0, and Q4_K_M GGUF builds, not the INT4 safetensors.

Ollama

Available on the [Ollama library](#).

Controlling reasoning

Laguna S 2.1-INT4 uses the same reasoning controls (interleaved thinking, preserved reasoning, and the `enable_thinking` flag) as the base model. See the [Controlling reasoning](#) section of the main Laguna S 2.1 model card.

License

This model is licensed under the [OpenMDW-1.1 License](#).

Intended and Responsible Use

Laguna S 2.1-INT4 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna S 2.1-INT4 is

subject to the [OpenMDW-1.1 License](#), and should be used consistently with Poolside's [Acceptable Use Policy](#). We advise against circumventing Laguna S 2.1-INT4 safety guardrails without implementing substantially equivalent mitigations appropriate for your use case.

Please report security vulnerabilities or safety concerns to security@poolside.ai.